Quicksort exercises

07-E01 Lomuto partition trace

Difficulty: Foundational.

Partition A=[7,3,8,3,6,2] with final element 2 as pivot. List every exchange and the final pivot index.

07-E02 A 1:9 recursion bound

Difficulty: Advanced.

Each partition costs n and the largest child has size at most 9n/10. Prove total O(n log n) by a recursion tree.

07-E03 Comparing four-rank endpoints

Difficulty: Advanced.

For four ordered distinct elements, analyze whether random Quicksort compares elements 1 and 4. Verify indicator probability 1/2.

07-E04 Equal keys and random pivots

Difficulty: Research.

All n keys equal 5. Lomuto sends keys $\leq$pivot left; each pivot is chosen randomly and moved to the end. Derive exact comparisons, identify the missing premise in "expected O(n log n) on every input," and give a duplicate-key repair.

07-E05 Smaller-side recursion stack

Difficulty: Research.

Recurse only on the smaller partition and continue the larger in a while loop. Prove worst-case O(log n) stack. Use an always-maximum pivot to show why time can remain $\Theta(n^2)$.

07-E06 Negative and repeated keys

Difficulty: Advanced.

Use 1-based Lomuto on A=[−1,4,0,4,2] with final pivot 2. List i,j and the array at each step, q, and the three final regions. Does partition imply sorted sides?

Work independently before opening the separate solutions PDF.

Quicksort exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study

Lin Cing
https://github.com/xinhi6716/computer-science-self-study